इंटरनेट मानक

Disclosure to Promote the Right To Information

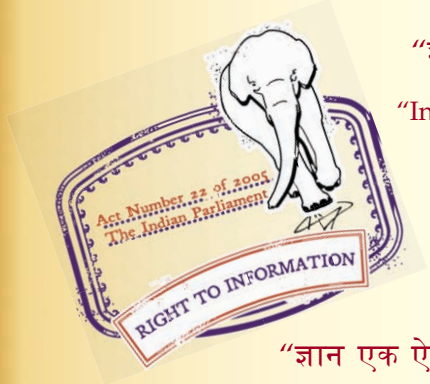
Whereas the Parliament of India has set out to provide a practical regime of right to information for citizens to secure access to information under the control of public authorities, in order to promote transparency and accountability in the working of every public authority, and whereas the attached publication of the Bureau of Indian Standards is of particular interest to the public, particularly disadvantaged communities and those engaged in the pursuit of education and knowledge, the attached public safety standard is made available to promote the timely dissemination of this information in an accurate manner to the public.

"जानने का अधिकार, जीने का अधिकार"
Mazdoor Kisan Shakti Sangathan
"The Right to Information, The Right to Live"

"पुराने को छोड नये के तरफ"
Jawaharlal Nehru
"Step Out From the Old to the New"

IS 10558 (2013): Aluminium Shore Gangways [TED 17: Transport Engineering]

"ज्ञान से एक नये भारत का निर्माण"
Satyanarayan Gangaram Pitroda
"Invent a New India Using Knowledge"





"ज्ञान एक ऐसा खजाना है जो कभी चुराया नहीं जा सकता है"
Bhartṛhari—Nītiśatakam
"Knowledge is such a treasure which cannot be stolen"

भारतीय मानक

एल्यूमिनियम शोर गैंगवे — विशिष्टि

(पहला पुनरीक्षण)

Indian Standard

ALUMINIUM SHORE GANGWAYS — SPECIFICATION

(First Revision)

ICS 47.020.50

© BIS 2013

BUREAU OF INDIAN STANDARDS
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI 110002

April 2013 **Price Group 4**

Shipbuilding Sectional Committee, TED 17

FOREWORD

This Indian Standard (First Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Shipbuilding Sectional Committee had been approved by the Transport Engineering Division Council.

This standard was first published in 1983. In this standard besides other changes, 'Regid-narrow-decking, two-handrail gangway' has been excluded as the same is no more used.

Ships require a means of ship-to-shore access which can be safely used by all persons. The aluminium shore gangway provides a lightweight convenient structure which can be used in a horizontal or inclined position up to an angle of 30° from the horizontal dependent on tidal or freeboard conditions. Special designs of step allow for a more severe angle to be employed than the 30° specified. The gangways described in this standard are not intended to carry wheeled traffic, such as, loaded trolleys, etc.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

ALUMINIUM SHORE GANGWAYS — SPECIFICATION

(First Revision)

1 SCOPE

This standard specifies requirements for two types of aluminium ship-to-shore gangways which are for the use of persons only to embark or disembark safely.

2 REFERENCES

The following standards contain provisions which through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards indicated below:

IS No.	*Title*
733 : 1983	Specification for wrought aluminium and aluminium alloy bars, rods and sections (for general engineering purposes) (*third revision*)
736 : 1986	Specification for wrought aluminium and aluminium alloy plate for general engineering purposes (*third revision*)
737 : 1986	Specification for wrought aluminium and aluminium alloy sheet and strip for general engineering purposes (*third revision*)
740 : 1977	Specification for wrought aluminium and aluminium alloy rivet stock for general engineering purposes (*second revision*)
1084 : 2005	Textiles — Manila ropes — Specification (*fifth revision*)
1284 : 1975	Specification for wrought aluminium alloy bolt and screw stock (for general engineering purposes) (*second revision*)
1285 : 2002	Wrought aluminium and aluminium alloys — Extruded round tube and hollow sections for general engineering purposes — Specification (*third revision*)
1321	Sisal ropes — Specification:
(Part 1) : 2003	Untarred varieties (*fourth revision*)
(Part 2) : 1982	Tarred varieties (*second revision*)
2062 : 2006	Hot rolled low, medium and high tensile structural steel (*sixth revision*)
3109 (Part 1) : 1982	Specification for short link chain, grade M4 : Part 1 Non-calibrated load chain for lifting purposes (*second revision*)
4009	Specification for grease nipples:
(Part 1) : 1981	Button head grease nipples (*first revision*)
(Part 2) : 1981	Conical head grease nipples (*first revision*)
(Part 3) : 1995	Cup type — Specification
4736 : 1986	Specification for hot-dip zinc coatings on mild steel tubes (*first revision*)
5175 : 1992	Polypropylene ropes (3-strand howser-laid and 8-strand-plaited) — Specification (*second revision*)

3 TERMINOLOGY

For the purpose of this standard, the following definitions shall apply.

3.1 Gangway — A bridge structure to allow safe embarkation and disembarkation from ship to shore or access to another ship.

3.2 Side Stringers — The longitudinal strength members of the gangway to which the cross-members, stanchions, roller or wheels, the lifting lugs, etc, are attached.

3.3 Cross-Members — The bars, angles or hollow sections which hold the side stringers in position, and provide support for the decking.

3.4 Decking — Flat topped corrugated section or plate.

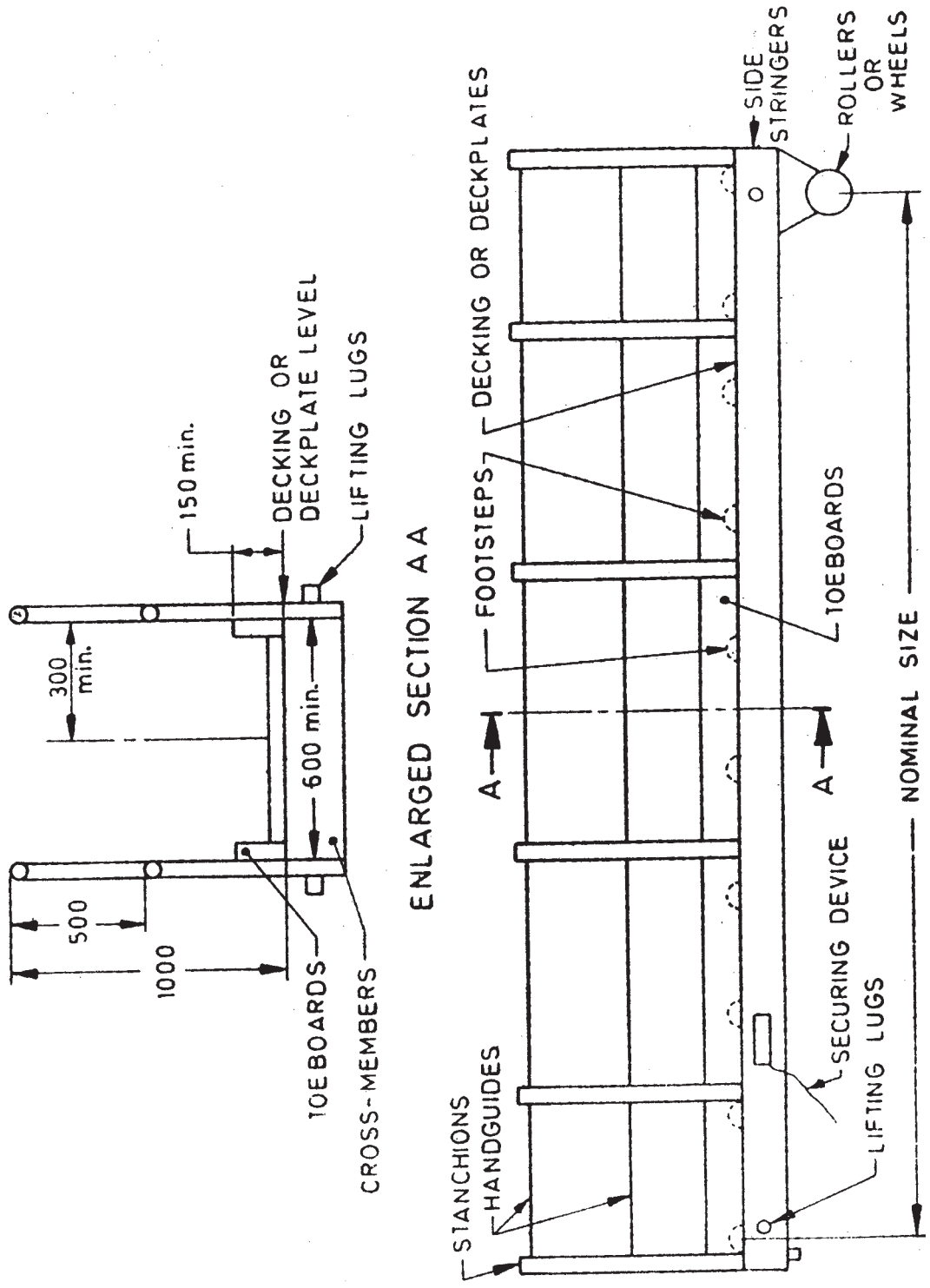
3.5 Footsteps — The bars, rounded hollow sections or hardwood battens fitted proud of the decking or deck plate level to give better foot grip when the gangway is inclined from the horizontal position.

3.6 Hand and Intermediate Guides — The tubes, bars, ropes or chains which protect people from falling from the gangway.

3.7 Nominal Size — The length of the gangway between points of support (*see* Fig. 1 and Fig. 2).

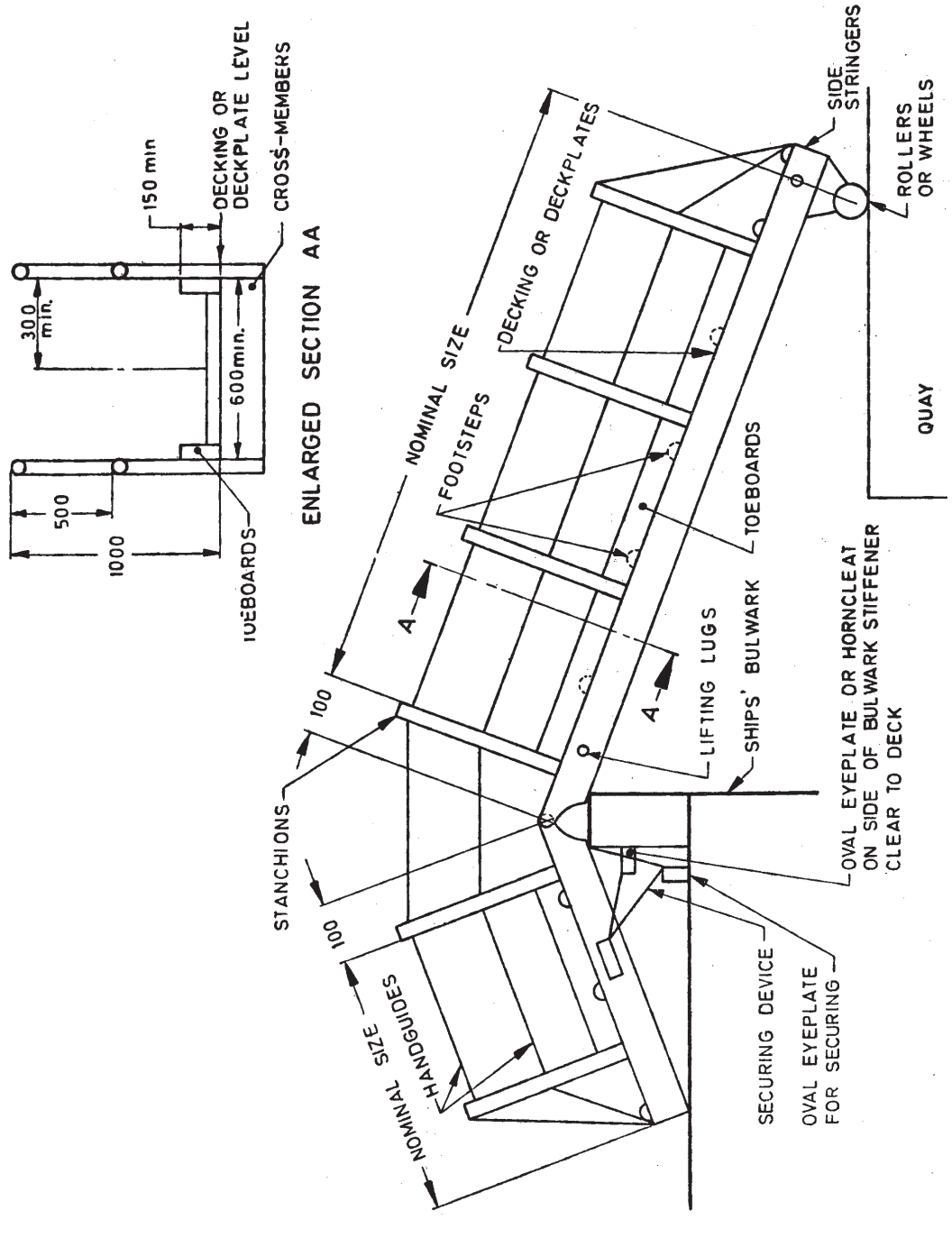
4 TYPES OF GANGWAY

The following two types of gangways are specified in this standard:



All dimensions in millimetres.

FIG. 1 GENERAL ARRANGEMENT OF TYPE A GANGWAY



NOTE — Height of the bulwark end stanchions to give minimum height to handguide of 1 000 mm.

All dimensions in millimetres.

FIG. 2 GENERAL ARRANGEMENT OF TYPE B GANGWAY

a) *Type A* — Rigid-decking two-handrail gangway (*see* Fig. l).

b) *Type B* — Hinged, two-handrail gangway (*see* Fig. 2).

NOTE — The longer Type A gangway with hinged or portable handguides may be made up of sections connected by rigid jointing.

5 MATERIALS

The materials shall comply with the requirements of Table 1. Other materials may be used provided they have equivalent strength and are compatible with the structural use of aluminium.

6 DIMENSIONS

6.1 Gangways shall be designed to the nominal sizes shown in Tables 2 and 3 for Types A and B, respectively.

6.2 The minimum width between side stringers shall be as indicated in Fig. 1 and Fig. 2.

7 DESIGN AND CONSTRUCTION

7.1 Design

7.1.1 The manufacturer of the gangway shall unusual or hazardous conditions affecting the design of the gangway.

7.1.2 The manufacturer shall ensure that,

a) where dissimilar metals are used in the construction of an aluminium shore gangway their direct contact shall be avoided, to prevent galvanic corrosion. Where aluminium and mild steel have common faying surfaces the mild steel is to be galvanized and the surfaces are to be etched prime with zinc chromate primer and coated with jeffery jointing compound. Connecting nuts, bolts and washers are to be dipped in jeffery jointing compound before being used;

b) each gangway is provided with an effective and marked earthing device;

c) the gangway is suitably surface coated to prevent incendive sparking; and

d) antiseize compound is applied to the thread of aluminium alloy bolts before screwing on nuts.

7.1.3 *Design Loading*

The assembly, comprising side stringers, cross-members and decking shall be designed to withstand a uniform deck loading of 4 000 N/m^2 applied to the decking and treads whilst the gangway is in a horizontal position.

7.1.4 *Factor of Safety*

The allowable stress used in the design of the gangway with the loading specified in **7.1.3** shall be determined by applying a factor of safety of 2 on the 0.2 percent proof stress of aluminium.

Table 1 Material
(*Clause* 5)

Sl No.	Component	Material	Relevant Indian Standard (s)	Remarks
(1)	(2)	(3)	(4)	(5)
i)	Side stringers	Aluminium	IS 733, IS 736, IS 737, IS 740 IS 1284, IS 1285	H 30 alloy
ii)	Cross-members	Aluminium	IS 736, IS 737 IS 1285	
iii)	Decking or deck plates	Aluminium	do	
iv)	Footsteps	Aluminium, Hardwood	IS 1285	
v)	Rollers or Wheels	Carbon steel, Carbon steel with solid types	IS 2062	Grade 43 A Ribbed or Flat tread
vi)	Stanchions	Aluminium Carbon steel	IS 733, IS 1285, IS 2062	H 30 alloy grade 43 A
vii)	Handguides:			
	a) Handrail	Aluminium section	IS 1285	See 7.2.6
	b) Handrope	Sisal or manila	IS 1321 (Part 1)	See Note under 7.2.6
		Polypropylene	IS 1321 (Part 2), IS 1084	
			IS 5175	
	c) Handchain	Steel	IS 3109 (Part 1)	Galvanized
	d) Handwire, plastic coated	PVC coated guard-wire rope		
	e) Securing device	Sisal or manila	IS 1321 (Part 1)	
		Polypropylene	IS 1321 (Part 2), IS 1084	
		Polypropylene	IS 5175	See Note under 7.2.6
		Staple fibre		
	f) Toeboards	Aluminium, Hardwood	IS 733	H 30 alloy

Table 2 Nominal Sizes and Stowage Dimensions of Types A and B (Fixed Handguides)
(*Clause* 6.1)

All dimensions in millimetres.

Sl No.	Nominal Size	Maximum Stowage Dimensions of Uncrated Gangways with Fixed Handguides					
		Maximum Overall Length		Maximum Overall Width		Maximum Depth (Fixed Handguides)	
		Type A	Type B	Type A	Type B	Type A	Type B
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	3 000	3 750	3 500	650	650	1 400	Made up from two nominal sizes. Use maximum depth dimensions for type A gangway for relevant sections
ii)	4 000	4 750	4 500	650	650	1 400	
iii)	5 000	5 750	5 550	650	650	1 500	
iv)	7 000	7 750	7 600	650	650	1 500	
v)	9 000	9 750	9 600	650	650	1 500	
vi)	12 000	12 750	12 600	800	800	1 600	
vii)	15 000	15 750	15 600	800	800	1 750	
viii)	18 000	18 750	18 600	800	800	1 900	
ix)	21 000	21 750	21 600	800	800	2 050	
x)	24 000	24 750	24 600	800	800	2 200	
xi)	27 000	27 750	27 600	800	800	2 400	
xii)	30 000	30 750	30 600	800	800	2 600	

7.2 Construction

7.2.1 *Side Stringers*

Side stringers shall be constructed from welded, extruded hollow or rolled sections, or from plate or from any combination of these.

7.2.2 *Cross-Members*

Cross-members attached to the side stringers shall be capable of supporting the deck or deck plates.

7.2.3 *Decking*

The decking shall comprise either continuous flat-topped longitudinal corrugated section or individual flat-plate surfaces, which may have a non-slip coating.

7.2.4 *Footsteps*

Footsteps of 50 mm × 30 mm hardwood shall be secured to the top of the decking at regular centre-to-centre intervals of not less than 400 mm. Alternatively, the footsteps may be constructed from extruded aluminium sections secured to the stringers of the decking, and raised above decking level by a minimum of 30 mm, in which case the minimum longitudinal length of flat decking between adjacent footsteps shall be 350 mm.

7.2.5 *Stanchions*

Stanchions shall be constructed from carbon steel or aluminium to a height that complies with Fig. 1 and Fig. 2.They shall be fitted at regular intervals along the gangway, with a maximum permitted interval of 1 500 mm. Stanchions and associated hand guides shall be designed for a side loading at the upper guide level of 500 N/m, without permanent deformation to stanchions or rigid hand guides when used. Stanchions shall be erected vertically with the gangway horizontal and shall be of one of the following types:

a) Fixed;
b) Hinged, with provision made to prevent inadvertant collapse; and
c) Portable, with securement to prevent accidental displacement from the socket or base support.

Galvanized stanchions where specified shall comply with the requirements of IS 4736.

7.2.6 *Hand and Intermediate Guides*

Hand and intermediate guides for the gangway shall be provided which comply with Fig. 1 and Fig. 2 and shall be of one of the following types:

a) Continuous and adequately tensioned sisal, manila or polypropylene or plastic covered wire-rope of 16 mm minimum diameter.
b) Galvanized steel chain provided with adequate means of tensioning.
c) Continuous rigid aluminium solid or hollow section for use with fixed or portable stanchions.
d) Continuous rigid aluminium solid or hollow section for the hand-guides of hinged stanchions, with the intermediate guides as in **7.2.6** (a).

NOTE — Polypropylene ropes shall be certified effective against actinic degradation for two years exposure in tropical conditions.

7.2.7 *Toeboards*

Toeboards, to a height of at least 150 mm above the decking level, shall be fitted at each side of the gangway.

7.2.8 *Roller or Wheels*

A roller or wheels of 100 mm minimum diameter shall be positioned at one end of the gangway. Rollers or wheels shall be self-lubricated or fitted with a lubricating nipple which complies with the requirements of IS 4009 (Part 1) or IS 4009 (Part 2) or IS 4009 (Part 3) and shall have guards provided for foot protection when the ship moves. The maximum angle of use of the gangway shall not prevent loss of contact between the roller and wheels with the contact surface.

7.2.9 *Securing Devices*

Suitable attachments shall be provided on both sides of the gangway in order to connect the securing devices (*see* Fig. 1 and Fig. 2).

7.2.10 *Lifting Lugs*

The gangway shall be provided with four lifting lugs securely attached to the stringers and positioned to produce a balanced lift.

7.2.11 *Angle of Use*

The construction of the gangway shall allow for usage at any angle of inclination from the horizontal plane to 30° from the horizontal.

8 WORKMANSHIP

8.1 The assembly, comprising side stringers, cross-members, and decking, together with all ancillary fittings, shall be visibly free from defects and distortion.

8.2 All components shall be free from exposed rough or sharp edges likely to cause injury.

8.3 Care shall be taken in the preparation, riveting, bolting or welding of aluminium structures to ensure that the permissible design stresses are not exceeded.

8.4 The requirements given in **7.1.1** and **7.1.2** shall be verified, where applicable.

9 ACCEPTANCE TESTS

9.1 The following tests shall be carried out at the manufacturer's works.

9.2 *Manufacturer's Type Test*

One gangway of each nominal size and type shall be tested in accordance with Annex A and a test certificate made available to the purchaser on request.

9.3 *Individual Test*

Each gangway shall be fully assembled with all fittings comprising the dead load (that is the fully erected gangway) and tested as follows:

a) Lifting by lifting lugs;
b) Measurement of initial sag (*see* Annex A); and
c) Preparation for stowage and checking of stowage dimensions.

10 INSPECTION

10.1 Gangways subjected to a type test shall be inspected to ensure that there are no signs of residual weakness or damage.

10.2 All gangways shall be visually checked to ensure that,

Table 3 Nominal Sizes and Stowage Dimensions of Types A and B (Hinged or Portable Handguides)
(*Clauses* 6.1 *and* 14.1)

All dimensions in millimetres.

Sl No.	Nominal Size	Maximum Stowage Dimensions of Uncrated Gangways with (Hinged or Portable Handguides)						
		Maximum Overall Length		Maximum Overall Width		Maximum Depth (Stanchions Removed or Folded Flat)		
		Type A	Type B	Type A	Type B	Type A		Type B
						No. of sections	Depth	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	3 000	3 750	3 500	650	650	1	300	
ii)	4 000	4 750	4 500	650	650	—	300	
iii)	5 000	5 750	5 550	650	650	—	317	Made up from two nominal sizes. Use maximum depth dimensions for type A gangway for relevant sections
iv)	7 000	7 750	7 600	650	650	—	350	
v)	9 000	9 750	9 600	650	650	—	400	
vi)	12 000	12 750	12 600	800	800	2	700	
vii)	15 000	15 750	15 600	800	800	—	800	
viii)	18 000	18 750	18 600	800	800	—	900	
ix)	21 000	21 750	21 600	800	800	3	1 000	
x)	24 000	24 750	24 600	800	800	—	1 100	
xi)	27 000	27 750	27 600	800	800	—	1 200	
xii)	30 000	30 750	30 600	800	800	—	1 300	

NOTES

1 If greater angles of inclination are required, the purchaser shall specify the maximum angle to the manufacturer.

2 Stowed lengths may be short or if rigid-joint extendable gangways are used. The preferred number of sections of Type A gangways assembled in this way is shown in col 7.

a) there is no distortion of the side stringers;
b) decking or deck plates are adequately secured;
c) roller or wheels revolve freely;
d) if applicable, the stanchions, hand and intermediate guides may be easily erected in position;
e) if applicable, any folding sections can be secured without damage;
f) removable fittings for rigid joints are identified and properly stowed when the gangway is dismantled; and
g) rating plate is affixed and correct.

11 DESIGNATION

The gangway conforming to this Indian Standard shall be designated as follows:

a) Nominal size;
b) Type;
c) Maximum permitted angle of use; and
d) Number of this Indian Standard.

Example:

Designation of an aluminium shore gangway of nominal size 12 000, Type A, limited to 30° angle of use shall be 'Aluminium Shore Gangway, 12 000, A, 30°, IS 10558'.

12 MARKING

12.1 Each gangway shall be permanently marked by means of a rating plate prominently displayed. The rating plate shall contain such information as is relevant to the gangway, including the following:

a) Manufacturer's name or trade-mark, type, number and serial number;
b) Number of this Indian Standard;
c) Type of gangway, that is, A or B;
d) Nominal size; and
e) Maximum permitted angle of use.

12.2 BIS Certification Marking

The gangways may also be marked with the Standard Mark.

12.2.1 The use of the Standard Mark is governed by the provisions of the *Bureau of Indian Standards Act*, 1986 and the Rules and Regulations made thereunder. The details of conditions under which the licence for the use of the Standard Mark may be granted to the manufacturers or producers may be obtained from the Bureau of Indian Standards.

13 PREPARATION FOR DESPATCH

13.1 Gangways shall be prepared for despatch with suitable protection to prevent damage in transit. Loose gear shall be well secured or packed in separate crates or boxes. The gangway shall be clearly marked with the manufacturer's and purchaser's reference number or other identification.

13.2 Any special packing requirements for the gangway shall be specified by the purchaser.

14 INFORMATION TO BE SUPPLIED BY THE PURCHASER

14.1 At the time of the enquiry or order the purchaser shall provide the manufacturer with the following information:

a) Gangway to IS 10558;
b) Type of gangway (*see* **4**);
c) Nominal size (*see* Tables 2 and 3);
d) For a Type A gangway, statement of preference for number of sections (*see* Notes 1 and 2 under Table 3);
e) Type of stanchions required (*see* **7.2.5**);
f) Type of hand and intermediate guides required (*see* **7.2.6**);
g) Whether to be fitted with rollers or wheels (*see* **7.2.8**);
h) Desired length of securing devices (*see* **7.2.9**);
j) Angle of inclination, if over 30° (*see* **7.2.11**);
k) Whether a test certificate is required (*see* **9**);
m) If additional tests are to be performed;
n) Any special stowage requirements for gangway fittings; and
p) Any special protection required for despatch (*see* **13.2**).

15 INFORMATION TO BE SUPPLIED BY THE MANUFACTURER

When specified by the purchaser, the manufacturer shall supply the following information at the tendering stage:

a) Arrangement drawing of gangway with principal dimensions and attachment details; and
b) Mass of gangway.

ANNEX A

(*Clauses* 9.2 *and* 9.3)

METHOD OF TEST FOR INITIAL SAG AND DEFLECTION OF GANGWAYS

A-1 Initial sag shall be determined by placing the gangway horizontally on supports positioned according to the nominal size. The gangway is fully assembled with all fittings comprising the dead load. Stretch a thin string or wire tightly between the two support points and, measure the maximum vertical distance *r* appearing between the horizontal line and the base of the gangway. The procedure is adopted for both side stringers and the average of the two readings r_1 and r_2 is taken as

$$\text{Initial sag} = \frac{r_1 + r_2}{2}$$

A-2 The deflection test shall be carried out immediately after the results of initial sag are determined. Apply, without shock, a uniform load equivalent to 4 000 N/m^2 to the longitudinal centre line of the decking. The load should be arranged from a selection of convenient sized sandbags or other material which will not damage the gangways and located at intervals of not more than 1 000 mm. Where the design incorporates individual decking plates a load equivalent to 4 000 N/m^2 shall be applied to each plate. The test load is applied for 15 min before the deflection movement of the gangway is measured.

The maximum movement of each side stringer X_1 and X_2 is measured between the string or wire stretched tightly between the support points and the base of the gangway as follows:

$$\frac{X_1 + X_2}{2}$$

Calculate the deflection as follows:

$$\frac{X_1 + X_2}{2} - \frac{r_1 + r_2}{2}$$

The maximum deflection shall not exceed the limit of nominal size divided by 75.

Bureau of Indian Standards

BIS is a statutory institution established under the *Bureau of Indian Standards Act*, 1986 to promote harmonious development of the activities of standardization, marking and quality certification of goods and attending to connected matters in the country.

Copyright

BIS has the copyright of all its publications. No part of these publications may be reproduced in any form without the prior permission in writing of BIS. This does not preclude the free use, in the course of implementing the standard, of necessary details, such as symbols and sizes, type or grade designations. Enquiries relating to copyright be addressed to the Director (Publications), BIS.

Review of Indian Standards

Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition by referring to the latest issue of 'BIS Catalogue' and 'Standards : Monthly Additions'.

This Indian Standard has been developed from Doc No.: TED 17 (721).

Amendments Issued Since Publication

Amend No.	Date of Issue	Text Affected

BUREAU OF INDIAN STANDARDS

Headquarters:

Manak Bhavan, 9 Bahadur Shah Zafar Marg, New Delhi 110002
Telephones : 2323 0131, 2323 3375, 2323 9402 *Website*: www.bis.org.in

Regional Offices: *Telephones*

Central : Manak Bhavan, 9 Bahadur Shah Zafar Marg, NEW DELHI 110002 — 2323 7617, 2323 3841

Eastern : 1/14 C.I.T. Scheme VII M, V. I. P. Road, Kankurgachi, KOLKATA 700054 — 2337 8499, 2337 8561, 2337 8626, 2337 9120

Northern : SCO 335-336, Sector 34-A, CHANDIGARH 160022 — 60 3843, 60 9285

Southern : C.I.T. Campus, IV Cross Road, CHENNAI 600113 — 2254 1216, 2254 1442, 2254 2519, 2254 2315

Western : Manakalaya, E9 MIDC, Marol, Andheri (East), MUMBAI 400093 — 2832 9295, 2832 7858, 2832 7891, 2832 7892

Branches: AHMEDABAD. BANGALORE. BHOPAL. BHUBANESHWAR. COIMBATORE. DEHRADUN. FARIDABAD. GHAZIABAD. GUWAHATI. HYDERABAD. JAIPUR. KANPUR. LUCKNOW. NAGPUR. PARWANOO. PATNA. PUNE. RAJKOT. THIRUVANANTHAPURAM. VISAKHAPATNAM.

Published by BIS, New Delhi